

Ex. No. : 14	Weather Reporting System
Date: 06-10-2025	

Aim

To implement a Weather Reporting System using ESP32 using DHT22 sensor

Algorithm

- Step 1: Initialize ESP32 and configure DHT22 sensor.
- Step 2: Read temperature and humidity values from the sensor.
- Step 3: Check for validity of sensor data.
- Step 4: Display readings on Serial Monitor.
- Step 5: Repeat periodically (every few seconds).

Source Code

Include Library: DHT Sensor Library for ESPX

```
#include "DHTesp.h"
```

```
#define DHT_PIN 15 // DHT22 data pin
```

```
DHTesp dht;
```

```
void setup() {
  Serial.begin(115200);
  dht.setup(DHT_PIN, DHTesp::DHT22);
}
```

```
void loop() {
  TempAndHumidity data = dht.getTempAndHumidity();
```

```
  Serial.print("Temperature: ");
  Serial.print(data.temperature);
  Serial.println(" °C");
```

```
  Serial.print("Humidity: ");
  Serial.print(data.humidity);
  Serial.println(" %");
```

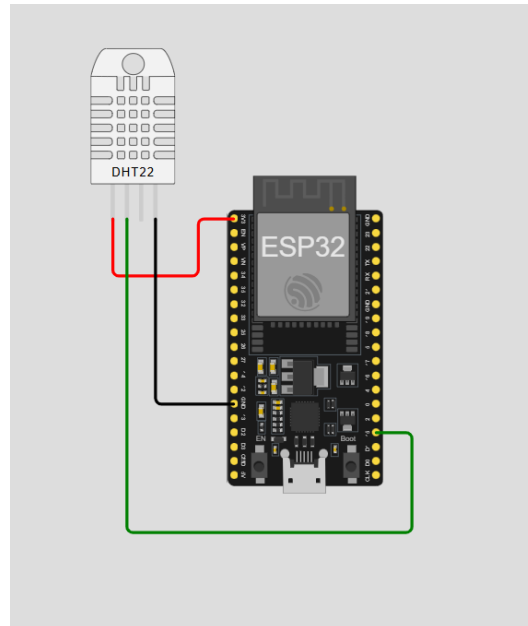
```
  if (isnan(data.temperature) || isnan(data.humidity)) {
    Serial.println("Failed to read from DHT sensor!");
  }
  Serial.println("-----");
  delay(3000); // update every 3 seconds
```

}

Circuit Connection

DHT22 Pin	Connection
VCC	3.3V
GND	GND
DATA	GPIO 15 (DHT_PIN)
Pull-up resistor 3.3V	10kΩ between DATA and 3.3V

Device Setup



Output

Temperature: 28.5 °C

Humidity: 66.3 %

Temperature: 29.1 °C

Humidity: 65.8 %

Result:

Thus the program has been executed and output verified successfully.